



Digital transformation requires digital resource primacy: Clarification and future research directions

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ABSTRACT

Responding to recent calls, this essay offers a commentary on the framing and definition of organizational digital transformation. We focus on the unique ontology of digital transformation and delineate it from neighboring concepts. Our contention is that, despite its volume, current research remains unclear about how the *digital transformation* of organizations differs from their *IT-enabled transformation*. We advocate definitional precision to foster knowledge accumulation and to enable scholars to pursue important research questions that are unique to digital transformation.

Our perspective, grounded in the notion of digital resources, defines digital transformation as the metamorphosis of an IT-enabled organization into a digital organization – one with a specific digital architecture and design principles. A key departure from previous conceptualization is that we characterize digital transformation as a change *in* digital technology architecture rather than a change *from* digital technology use. Our paper achieves the following: describes the constructs underpinning this formulation, digital resources and digital organization; justifies their use; and describes what research directions the new perspective promotes. With sound definitions of key constructs, Information Systems scholars have the unprecedented opportunity to lead the way in digital “x” research, making our discipline the reference point for the burgeoning “digital research” literature in related business fields.

More and more companies looking to stay relevant and compete in the digital age are attempting to become software businesses. It's an extremely challenging shift to tackle. Leaders can engage the effort with a transformation mindset and a commitment to make significant changes across the organization.

- McKinsey Insight (Gnanasambandam et al., 2022)

Introduction

This article answers the recent call “to clearly define digital transformation in a way that differentiates it from its neighboring concepts” (Markus & Rowe, 2023, p. 329). Clear definitions are the foundation of sound theoretical development and “are essential for

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the integration of knowledge” (Markus & Rowe, 2023, p. 331). The point of departure of our work is the belief that, as hinted in the opening quote, digital transformation is not the same as the IT-enabled transformation the Information Systems (IS) discipline has studied since the 1990s.¹ The latter involves socially curated packages of hardware and software that are adopted, installed, and used by employees and customers to achieve some desirable organizational outcome. It emerges from studies of the deployment of hardware and software assets that are conceptually separate from other organizational resources. Such a perspective lingers with many managers who, while understanding the importance of digital technology, “still tend to look at software as a [separate] capability that they can bolt onto their existing business models” (Gnanasambandam et al., 2022, p. 2).

We concur that human language does not remain static over time, with words assuming new meaning (Markus & Rowe, 2023). However, in scholarly endeavors, the introduction of new terminology should imply the material difference of new concepts from established ones (Leonardi, 2010). This is particularly important for IS scholars, given the ever-expanding role that digital technology assumes in organizations and society (Baskerville et al., 2020). As Faulkner and Runde (2019) recently put it, “Ignoring ontological considerations...limits the scope to account for properties things possess by virtue of their mode of being and for theorizing the relationship between ontologically distinct kinds of entity” (p. 1283). Thus, embracing the unique nature of digital transformation provides a wealth of opportunities to produce the kind of theory needed to make IS literature a reference for studying digital “x” phenomena (Baiyere et al., 2023).²

Recent research shows that digital innovations are ontologically different from traditional industrial innovations (Lyytinen, 2022) and that the nature of digital resources differs from that of traditional IT and IT-enabled resources (Piccoli et al., 2022). The same clear demarcation has yet to be established for digital transformation (Markus & Rowe, 2023). “IT boxes” are increasingly relegated to the background (Tilson et al., 2010; Yoo, 2010), suggesting that digital transformation hinges on the pliability of digital resources (Piccoli et al., 2022) to offer infinite configurations of value through multiple anticipated and unexpected value paths (Henfridsson et al., 2018).

With respect to the object of transformation, our focus is on the organization.³ Thus, we do not address change in super-organizational entities (e.g., markets, organized society), and we do not focus on the *process of transformation* itself. In line with Markus and Rowe (2023), we conceptualize transformation as a metamorphosis, a “qualitative or step-functional difference” (p. 331) between an organizational beginning and end state. Finally, with respect to our view of “digital,” we do not emphasize either a particular type of technology (e.g., SMACIT⁴ vs. ERP) or a magnitude change in data or processing power. Rather, we emphasize the unique ontology of digital objects and digital resources as the catalyst for digital transformation. We contend that embracing the strategic primacy of digital resources requires transformational change in organizational deep structures (Besson & Rowe, 2012).

Research in this area is important given the potential impact of successful digital transformation for organizations. Recent academic literature documents the superior growth potential of “inverted firms” – those companies that successfully deploy digital resources that are leveraged into value-creating initiatives by complementors (Benzell et al., 2022). Firms drawing on digital infrastructures scale their business at unprecedented rates (Giustiziero et al., 2023) and can benefit from data-driven operations, instant release practices, and swift transformation skills (Huang et al., 2017). Beyond sheer growth and financial success, digital transformation is expected to contribute to organizational agility (Salmela et al., 2022), responsiveness, and speed to market (Kaganer et al., 2023). Traditional companies that embrace a modular business architecture grounded in digital resources reduce complexity, improve coordination, and consequently, become more agile (Greeven et al., 2021). Practicing managers and consultants also point to the opportunities available to organizations that embrace innovation in the creation, reuse, and recombination of digital resources as a key driver of their value proposition and internal operations (Gnanasambandam et al., 2022; Lawson, 2021; Yefim et al., 2021).

Unfortunately, however, most literature seems to implicitly treat organizational digital transformation as a type of, or equivalent to, IT-enabled transformation (Vial, 2019) and generally describes it as a form of organizational change induced by digital technology. This mere change in nomenclature involves swapping terms (*IT* to *digital*) while still largely describing an IT-enabled phenomenon. We contribute to the discourse by first clarifying the difference between the pre- and post-transformation organizational states. In doing so, we focus on architectural changes and design principles, while singling out *digital resources* as the distinctive element at the core of the transformation to a *digital organization*.

The paper proceeds by first summarizing the central themes of extant digital transformation definitions. We then describe the core concepts in our theorizing, namely digital resources and digital organization, leading to a definition of digital transformation as the *metamorphosis of an organization from one that is enabled by IT to a digital organization characterized by the strategic primacy of digital resources*. Finally, we draw implications from our thesis for organizational and business scholars and offer directions for future research.

¹ The literature has historically used the terms IT transformation (Gregory et al., 2015), IT-enabled transformation (Wessel et al., 2021) and IS-enabled transformation (Besson & Rowe, 2012) somewhat interchangeably. They connote a type of organizational transformation that features digital technology as a key enabler. We refer to them collectively as IT-enabled transformation.

² In line with previous work, when referring to “x” phenomena we mean “any past IS research stream’s pivotal categories” (Baiyere et al., 2023, p. 463), such as innovation, transformation, strategy, or infrastructure.

³ Much of the literature we leverage, and our own case work, focuses on for-profit ventures. However, our objective is to provide a general definition of digital transformation, one that will apply to all forms of organizing. In this paper, we use the term “organization” when making general statements and the terms “firm,” “company,” and “venture” as synonyms for “organization” when citing specific literature or examples that warrant the latter terminology.

⁴ Common acronym for social, mobile, analytics, cloud, and Internet of Things (IoT) technologies.

Digital transformation literature

Over the past decade, academic literature has examined the topic of digital transformation. Following a surge in both the topic's prominence and the volume of related research papers, recent studies have endeavored to synthesize accumulated knowledge and refine our understanding (Hanelt et al., 2021; Schallmo et al., 2017; Vial, 2019). While some works focus on digital transformation in isolation (Verhoef et al., 2021; Zhu et al., 2021), others juxtapose it against neighboring concepts (e.g., IT or IT-enabled transformation) from earlier literature (Gong & Ribiere, 2021; Hanelt et al., 2021; Schallmo et al., 2017; Vial, 2019). Despite these attempts, many scholars lament the lack of progress in delineating the unique characteristics of digital transformation (Markus & Rowe, 2023). An analysis of the extant literature reveals the following claimed distinctions between IT-enabled and digital transformation:

- **Difference in scope of change:** IT-enabled transformation targets the business processes internal to an organization while digital transformation has a more expansive reach. Digital transformation is a process designed to “improve an entity by triggering significant changes to its properties through combinations of information, computing, communication, and connectivity technologies” (Vial, 2019, p. 118). This distinction pertains to the *scope of change* from a narrow focus on business processes to broad attention to a set of organizational properties, such as product innovation, or to include changes in wider ecosystems and industrial sectors (Cennamo et al., 2020; Drechsler et al., 2020; Hanelt et al., 2021; Kaganer et al., 2023; Singh & Hess, 2017; Soluk & Kammerlander, 2021).
- **Difference in strategic level of change:** IT-enabled transformation involves *traditional back-end processes* while digital transformation is the use of digital technologies to *transform strategy and business models* by leveraging data-driven processes, ecosystems, and platforms (Cennamo et al., 2020; Drechsler et al., 2020; Hanelt et al., 2021; Kaganer et al., 2023; Singh & Hess, 2017; Soluk & Kammerlander, 2021). The main distinction lies in the strategic *impact* of the transformation: from a model anchored in operational efficiency, typically confined to the domain of IT departments, to a comprehensive approach that seeks to redefine the mechanisms of value creation and appropriation across the entire organization (Gregory et al., 2015).
- **Difference in identity or value proposition:** IT-enabled organizational transformation activities leverage digital technology in *supporting the firm's existing value proposition* and result in the *enhancement of an existing organizational identity*. Digital transformation activities leverage digital technology in *(re)defining an organization's value proposition* and involve *the emergence of a new organizational identity* (Soluk & Kammerlander, 2021; Wessel et al., 2021). This distinction pertains to changes in the organization and its identity from enhancement of current identity to a redefinition of such identity.
- **Difference in digital technologies used:** IT-enabled transformation is rooted in the use of traditional enterprise systems, such as ERP, whereas digital transformation is triggered and shaped by the widespread diffusion of contemporary digital technologies, such as cloud, social media, and big data analytics (Hanelt et al., 2021; Schallmo et al., 2017; Sebastian et al., 2017; Singh & Hess, 2017; Soluk & Kammerlander, 2021; Warner & Wäger, 2019). This distinction hinges on the *novelty* of digital technology enabling change and delineates a distinction between “modern” and “legacy” IT, claiming that digital transformation leverages the characteristics of “modern” digital technology while IT-enabled transformation predominantly revolves around “legacy” systems.

In essence, the literature indicates that digital transformation relies on the use of emerging digital technologies to influence organizational processes and to transform business models and strategy. It suggests that digital transformation results in a different organizational identity and instigates profound changes within the organization and its broader ecosystem. But if we assume that today's cutting-edge digital technologies, such as social media, cloud, and platforms, will become tomorrow's legacy systems, then the distinction primarily hinges on the *current impact* of these technologies. This stance relegates the difference between IT-enabled and digital transformation to a narrow level based on the latter somehow having a greater impact on the organization. Yet, it gives us no novel theoretical foundation since earlier research already shows how deep structure changes are the hallmark of IT-enabled transformations, and indeed of all business transformations (Besson & Rowe, 2012).

Perhaps more fundamentally, a focus on technology impact, instead of its design, confounds the appropriate unit of analysis, and it fails to identify structural and ontological differences that would warrant the use of a new term – digital viz IT-enabled. Both IT-enabled and digital transformations occur within individual organizations or in collaboration with partner networks (Cennamo et al., 2020; Rai et al., 2006). Emphasis on the consequences of organizational transformation, such as identity or business model change, does not clarify the uniqueness of digital transformation. This makes it difficult to search for the root catalyst of the transformation, a prodromic step to theorizing the transformation's process and its effects. Without clear demarcation, the introduction of a new term – “digital” transformation – is unwarranted and misleading (Markus & Rowe, 2023). Attesting to the confusion, recent work even suggests that digital transformation “is not different, ontologically speaking, from IT-enabled transformation” and that we should disregard labels and “focus instead on the interesting and novel features of both” (Carroll et al., 2023, p. 8). We respectfully disagree. The digital transformation concept has a unique ontology that sets it apart from any other kind of transformation. What is missing in the literature is a first-principled analysis of those recognizable changes of material import (Leonardi, 2010) that make digital transformation a unique phenomenon.

Digital resources and the digital organization

The *Britannica Dictionary* defines transformation, when applied to an entity rather than a process, as “a thorough or dramatic change in form or appearance.” The *Merriam-Webster Dictionary* provides a similar description: “change of physical form, structure, or substance.” Thus, while the liminality of transformation processes can be discrete or ongoing (Haskamp et al., 2022), an organizational

transformation implies a metamorphosis, from the Greek μετά (meta) “change,” and μορφή (morphē), “form.” A transformation requires the entity’s conversion from a beginning state to a recognizably different end state. Consider the three changes of state of the H₂O compound when the temperature rises: ice melts into water, and water evaporates into steam. Such an end state may be a stable long-term equilibrium or a transitory condition, with a linear or non-linear transformation process bridging the two (e.g., the change of state from ice to water is a transitory transformation in the above example). Regardless of the process, though, a significantly altered end state is definitionally required to identify the “change in form, structure, or substance” that qualifies a *transformation*.

Of course, a transformation is not the only type of organizational change. A firm can evolve in several dimensions over short- or long-term horizons. It can grow its customer base or narrow its product scope; it can vertically integrate or divest non-strategic units; it can improve some of its capabilities or acquire new assets. Definitionally, none of these variations necessarily configures as a transformation unless a significant *change in form or structure* is evident after the change has run its course.

Recent literature embraces the conceptualization of digital transformation as a metamorphosis (Baiyere et al., 2020; Markus & Rowe, 2023). However, no work to date has clearly delineated the characteristics of the end state as materially different from the beginning state of a digital transformation. Such failure makes it difficult to identify the specific changes in structure or substance that characterize the unique nature of digital versus IT-enabled transformations (Markus & Rowe, 2023). We contend that identity change, value proposition change, and organizational change, despite their prevalence in the literature, are neither necessary nor sufficient elements of a digital transformation. For example, BMW stopped making plane engines in 1918 and began to manufacture cars. A significant change of identity and value proposition took place. But in 1918, it clearly did not imply or stem from a digital transformation. Amazon.com, Inc. is the prototypical early example of digital transformation, having started in 2002 to reconfigure its monolithic digital technology core into a collection of digital resources (Burgin & Orosz, 2023). Despite such digital transformation, at the time, Amazon manifested no value proposition adjustment or any evidence of identity change (Stone, 2013). The juxtaposition of these two examples shows that while “a thorough and dramatic change” of identity, value proposition, or organizational structure may be associated with organizational transformation (e.g., BMW’s dramatic change of product lines), they are neither necessary nor sufficient characteristics of *digital transformation*.

We resolve the tension by proposing the following first-principled definition: A digital transformation is *the metamorphosis of an organization from one that is enabled by IT to a digital organization characterized by the strategic primacy of digital resources*. This definition highlights the deep structural change in the architectural design of information technology resources underpinning the firm’s operations and value proposition, and implies the proactive management of the accompanying systemic effects (Bostrom & Heinen, 1977). The transformation does not occur *because* of technology, as a technocentric view would imply. Rather, as with any socio-technical phenomenon, change stems from the synergic design of technology, organizational structure, people, and processes (Bostrom & Heinen, 1977). Yet, the pivotal aspect of the transformation is the firm’s *strategic choice* (Van den Steen, 2017) to *commit to digital resources as the core element of its digital technology architecture*. Importantly, embracing the strategic primacy of digital resources is not the mere use of “modern” digital technology (e.g., SMACIT vs ERP). Rather, it is a thorough and dramatic change in the design principles driving the firm’s digital technology choices – spanning the firm’s infrastructure, its value creation processes, and its value propositions (i.e., products and services). Note that this definition applies just as well to firms offering digital products (e.g., Netflix) as to industrial stalwarts producing physical products that have not changed in decades (e.g., Tetra Pak). In the latter case, the change reflects in infrastructure, business activities, or industrial processes – not necessarily in the products.⁵

Like any form of organizational transformation, digital transformations imply changes in the “configuration of the formal organizational arrangements, including the formal structures, processes, and systems that make up an organization” (Nadler & Tushman, 1997). Yet, the focus on technology architecture as the structural driver of digital transformation is important to highlight the defining nature of digital transformation. In the remainder of this section, we justify our definition, explaining the distinctive structure of the end state, a *digital organization*, and the technological foundations at its core: *digital resources*.

Digital resources⁶

Digital resources are “a specific class of digital objects that a) are *modular*, b) *encapsulate objects of value*, assets and/or capabilities, c) and are accessible by way of a *programmable interface*” (Piccoli et al., 2022, p. 2293). Digital resources have unique structural characteristics. First, they are a type of digital object: a “structured continuant ... whose component parts include one or more bitstrings” (Faulkner & Runde, 2019, p. 1285), where bitstrings are “the sequences of 1 s and 0 s used in computing to represent information in binary form” (Faulkner & Runde, 2019, p. 804). Digital resources can configure as nonmaterial or as hybrid digital objects and always have a programmable bitstring interface.⁷ Second, they encapsulate an organizational object of value, either an asset or a capability. Thus, digital assets are digital resources that encompass either nonmaterial digital objects like a catalog of digital songs, or hybrid digital objects, like a virtualized computing resource (e.g., AWS S3), a smart device (e.g., Amazon’s Echo), or a self-driving vehicle (e.g., Caterpillar, Inc.’s 789D mining truck). Digital capabilities are repeatable patterns of organizational actions (Helfat &

⁵ The definition applies to governmental institutions (such as the tax authority) or non-profit organizations (such as universities). In these contexts, while speaking of “societal impact” rather than “value proposition” or “guiding principles” rather than “value creation processes,” the substance of the definition does not change.

⁶ For a comprehensive treatment of the digital resource concept, we defer to Piccoli et al. (2022).

⁷ APIs are specifications – in the form of routine definitions, protocols, and tools – for programmatic access to an application functionalities or data. APIs are the most popular approach, albeit not the only one, for implementing the interface of digital resources.

Raubitschek, 2018), yielding the capacity to undertake activities that a firm can access programmatically through a digital interface (e.g., Stripe Payment).

Note that digital resources are structured as modular components,⁸ enforcing information hiding and therefore ensuring that they can interact with the other elements in the system's design architecture strictly through their interface (Schilling, 2000). Their design as modular components ensures that they are systematically reusable and recombining by external orchestrators through their programmatic interface (Piccoli et al., 2022). Finally, digital resources are designed for reuse and recombination strictly through their programmatic interface rather than a manual or physical interface like most IT resources. Thus, they are visible and usable exclusively as digital abstractions – irrespective of whether the resource is internally a nonmaterial or hybrid digital object.

From a strategic standpoint, the ontology of digital resources implies that they can be reused and recombined in novel value paths (Henfridsson et al., 2018) through their programmatic bitstring interface. The unique nature of digital resources spurred theorists to suggest that “conceptualizing digital resources as resources that exhibit modular design, encapsulated value, and a programmatic interface may allow for clearer distinctions between IT-enabled transformation and digital transformation” (Markus & Rowe, 2023, p. 332). In the next section, we follow that logic to characterize the end state of digital transformation: a digital organization.

Digital organizations

Organizations can be conceptualized as bundles of resources (Penrose, 1959). According to the Resource Based View (RBV) of the firm (Barney, 1991; Peteraf, 1993; Wernerfelt, 1984), resources are “those (tangible and intangible) assets that are tied semi-permanently to the firm” (Wernerfelt, 1984, p. 172) and form “the basis for competition among firms” (Nevo & Wade, 2010, p. 164). Surprisingly, while several articles use terms like digital organization, digital firm, or digital native organization, there is no clear definition or consensus on what a digital organization is. As an example of the lack of attention to the unique nature of digital phenomena, the bulk of papers point to the use of software or the selling of digital products as the defining characteristics of these “digital” organizations (see Table 1).

We offer a rigorous definition grounded in the core guidance perspective of strategy. We conceive of digital organizations as those that ascribe *strategic primacy to digital resources*, thus prioritizing digital resources in their resource bundle. Digital organizations make the strategic choice to commit to the creation, reuse, and recombination of digital resources as a primary driver of their value proposition and internal operations. The term *strategic choices* here refers to those high-level decisions that comprise the organization's core guidance. In the strategy literature, the core guidance perspective views “strategy as the smallest set of choices to optimally guide (or force) other choices” (Van den Steen, 2017, p. 2616) to ensure the coherence of decisions and actions across all hierarchical levels and to minimize instances where “the organization's actions – although maybe sensible in isolation – do not fit together” (Van den Steen, 2018, p. 576). Recognizing the strategic primacy of digital resources, the digital organization will prioritize their use in its infrastructure, its value creation processes, and its value propositions. As a strategic choice rather than a technical decision, embracing the primacy of digital resources precipitates deep structural change in the firm's digital architecture and complementary adjustments in organizational structure, talent development and retention, firm culture, and ecosystem positioning.

Amazon's digital transformation, initiated in 2002, provides the canonical example of this functional definition in action. Realizing that the technical debt (Woodard et al., 2013) accumulated in the monolithic IT infrastructure underpinning its e-commerce operations hampered the firm's agility and execution speed (Burgin & Orosz, 2023), the CEO Jeff Bezos issued a mandate: No new development and no new initiatives would be approved unless they created or used “business primitives.” Amazon always considered itself a technology company, but the executive team realized that the firm's digital technology architecture and the *design* of the firm's technology resources were as important as their *use*. The firm stopped all new development and embraced the strategic primacy of digital resources by rearchitecting existing organizational resources and reorganizing the way they would be deployed with the goal of maximizing option value (Woodard et al., 2013), speed of execution, and agility (Stone, 2013). From a technology and organizational design standpoint, this transition implied a dramatic change in the architecture of resources such that the firm needed to “break its infrastructure into the smallest, simplest atomic components and allow developers to freely access them with as much flexibility as possible” (Stone, 2013, p. 162).

As a consequence of its digital transformation, organizational interactions at Amazon increasingly relied on digital resource orchestration. Commitment to this choice was famously expressed by Bezos' own dictate that small product teams expose their data and functionality exclusively through APIs (Benzell et al., 2022). Teams did not coordinate manually to use their resources, and all interaction among these resources occurred exclusively via programmatic digital interfaces (Lardinois, 2011). Upon enforcing the mandate requiring the creation and use of “business primitives,” the firm's digital architecture was redesigned; the very fabric of the firm was built upon digital resources, and future growth would follow the same design principles. Interestingly, Amazon executives realized the potential of the new digital architecture for other firms, particularly startups, and launched Amazon Web Services (AWS) in 2006. AWS today is a collection of over 200 digital resources⁹ that can be self-provisioned and governed programmatically with no

⁸ The term “modular” can be confusing when referring to a system's component rather than the overall system's architecture. In this paper, the term does not imply that the module has an internal modular architecture. Instead, it means that the “modular component” performs as a module within a larger system architecture (Schilling, 2000).

⁹ <https://www.aboutamazon.com/what-we-do/amazon-web-services>.

Table 1
Sample definitions for digital organizations.

Definition	Conceptual Challenges	Theme
Digital organization “a firm whose value offering comprises a significant digital or data-enabled component.” (Salmela et al., 2022, p. 1081)	While digital or data-enabled components are conceptualized as essential resources of digital organizations, they are not formally defined. The implication is that it is unclear what the distinction is between “digital or data-enabled components” and IT or IT-enabled resources of the kind organizations have been using for decades.	Selling digital value propositions
Digital firms , contrasted to “industrial firms,” are defined as “firms that participate heavily in the digital economy by either using a significant share of digital resources (e.g., software, algorithms, data) and/or by selling a significant share of digital products and services (e.g., platforms, software, media).” (Giustiziero et al., 2023, p. 1392)	Despite digital resources being conceptualized as a defining element of digital firms, their nature is only marginally addressed as “essentially scale free” resources. Examples of digital resources are “software, algorithms, and data,” which are traditionally considered IT resources. Moreover, digital products are not differentiated from IT-enabled ones. The result is that digital firms appear to be no different from traditional IT-intensive organizations of the kind the IS literature has studied for decades. Digital organizations are differentiated from pre-digital ones in terms of the timing in which they were incorporated (i.e., pre- and post-digital economy). Using the timing of incorporation as a criterion to define pre-digital organizations fails to capture their structural characteristics and ontology. Additionally, discerning any structural distinction among digital, IT-enabled, and non-digital organizations solely based on their creation date becomes challenging as companies in different industries, markets, or geographies will utilize traditional IT resources or digital resources regardless of when they were established.	Leveraging software and algorithms Selling digital value propositions
“ Pre-digital organizations are established companies belonging to traditional industries, such as retail, automotive, or financial services, that were financially successful in the pre-digital economy, but to which the digital economy poses an existential threat.” (Chanias et al., 2019, p. 17)	Despite identifying digital infrastructure as a distinctive attribute of digital organizations, the concept is not defined. Moreover, the distinction between digital infrastructures and IT infrastructures is not addressed. Considering that IT infrastructures have facilitated online commerce and electronic government by leveraging IT and IT-enabled resources for decades, the demarcation between digital and IT infrastructure, and digital and IT-enabled organizations, is ambiguous.	Timing of creation or incorporation
A digital firm “builds and leverages digital infrastructure” “to accrue...capabilities that allow the firm to both create and sell its offering online through a digital business model.” (Monaghan et al., 2020, p. 12)	Considering that digital artifacts can be configured as digital, IT, or IT-enabled resources, it remains unclear how, if at all, digital ventures (as defined here) differ from IT-enabled ones. Leveraging the concept of digital artifacts poses a significant challenge in distinguishing between digital and IT-enabled organization since both can incorporate digital artifacts as the core of their value propositions.	Leveraging hardware/software infrastructures Selling digital value propositions
Digital ventures are organizations “that have at the core of their market offerings a new digital artifact.” (von Briel et al., 2018, p. 278)	Digital market offerings are conceptualized as those that leverage digital technology components. Examples of digital market offerings include digital platforms or mobile apps. Despite digital technology components being critical to enabling digital market offerings, their nature is not addressed. The implication is that any product or service that incorporates any type of digital technology is considered a “digital market offering,” particularly in the hands of a “young firm,” making the organization a digital venture. There is no discussion on how digital market offerings differ from IT-enabled products or services.	Leveraging digital artifacts Selling digital value propositions
Digital ventures are “professionally funded and privately owned young firms that bring about digital market offerings.” (Lehmann et al., 2022, p. 1453)	Digital ventures are not formally defined. While leveraging a digital infrastructure is identified as a core attribute of these organizations, the concept is treated interchangeably with IT infrastructures. Therefore, it is not clear what is unique about digital organizations and how they differ, if at all, from IT-enabled ones.	Leveraging software resources Selling digital value propositions
“ Digital ventures grow by drawing on and adding to digital infrastructures.” (Huang et al., 2017, p. 301)		Leveraging software infrastructures

need for customers to coordinate with AWS staff (Piccoli et al., 2022).¹⁰

Skeptics may challenge the notion of digital resources “primacy” as ambiguous, arguing that it is a surface-level, top-down declaration missing the necessary nuance for a complex topic like digital transformation. We submit instead that, when interpreted considering the core guidance definition of strategy, the concept is well specified. When issuing core guidance, the strategist formulates and announces a series of strategic choices that guide all subsequent decisions in the firm (Van den Steen, 2017). Committing to the strategic primacy of digital resources implies adherence to a very specific set of design principles that – if the transformation

¹⁰ Note that the launch of AWS in August 2006 was a serendipitous byproduct of the digital transformation that Amazon had successfully completed a few years earlier.

process is completed successfully – a results in visible technical and organizational changes. Strategic choices of this kind, far from being surface level, represent the way in which a large organization demonstrates intentionality and can coordinate action across hundreds or even thousands of employees to achieve deep structure changes (Besson & Rowe, 2012). In practice, then, ascribing strategic primacy to digital resources means that *the firm will orchestrate digital resources when available, create them when not available, and resort to employing non-digital organizational resources, such as physical assets, labor, or traditional IT resources, when necessary*. On average, digital resources become increasingly prevalent in the products, services, or infrastructure of the digital firm. Of course, actual deployment remains constrained by the feasibility of architecting resources as digital at any point in time.¹¹ For example, Caterpillar will still produce articulated trucks and large excavators, increasingly autonomous and networked, and Rio Tinto International Holdings Ltd. will purchase them to extract ore in its mining operations. Yet, while the ore remains a physical product, Rio Tinto can manage its equipment fleet as a collection of digital assets and orchestrate them as part of its digital initiatives (Piccoli et al., 2022). Unlike an IT-enabled organization, because of its core guidance and the design of its digital architecture, a digital organization will systematically prioritize digital resources, likely attaining superior option value (Woodard et al., 2013). For example, while many insurance companies are evaluating Large Language Models (LLM) to improve productivity in their customer service operations (Brynjolfsson et al., 2023), only those configured as digital organizations have the architecture, processes, competences, and structure to rapidly incorporate OpenAI's GPT-4 via its API, not as a one-off project but as a new digital resource within their digital architecture. Note that this is not synonymous with: the firm will “create and sell its offering online through a digital business model” (Monaghan et al., 2020, p. 13) or “participate heavily in the digital economy” (Giustiziero et al., 2023, p. 1392). A food packaging manufacturer (e. g., Tetra Pak) will not digitize its milk packaging materials, even after embracing the primacy of digital resources. However, even when physical products do not change, industrial firms can redesign their digital infrastructure, internal operations, and/or their value propositions.

Recent literature documents the wide-ranging impact of digital architecture choices, showing how they lead to a transformation in a firm's organizing logic (Sandberg et al., 2020). The shift toward the digital organizational form exemplified by Amazon has not been explicitly described in the academic literature, but related terms like “platformization” (Poell et al., 2019), API strategies (Iyer & Subramaniam, 2015), digital infrastructure (Tilson et al., 2010), IT architecture flexibility (Mikalef et al., 2021), firm inversion (Benzell et al., 2022), and designing for digital (Ross et al., 2019) are discussed. As a testament to the timeliness of the research we advocate here, leading management consulting firms promote “flexible tech architectures,” relying on core components like “APIs that can access the underlying data, algorithms, and processes in legacy systems; a set of microservices ... and a common data platform that knits disparate data sources” (Gnanasambandam et al., 2022, p. 5). To support this transition while remaining relevant, traditional IT providers like enterprise systems vendors are rearchitecting their systems with the so-called MACH architecture to offer “composable” solutions (Faith et al., 2020).¹²

We argue that grounding the notion of digital organization in the digital resources concept yields ontological precision, ensuring that the concept is rigorously defined and meaningfully differentiated from that of an IT-enabled organization. A digital organization may prioritize the use of digital resources by engaging in the internal reconfiguration of organizational resources (as in the Amazon example). Alternatively, it may primarily rely on an external software ecosystem like the Australian “cloud-based bank” Judo Bank (Breidbach et al., 2022) or Instacart (Li et al., 2022), a grocery delivery firm that assembles its value proposition by orchestrating digital resources from ecosystem partners like Stripe, Twilio, and AWS. Most often, a digital organization deploys a blend of internally developed and externally acquired digital resources, as in the case of Banco Bilbo Vizcaya Argentaria, a Spanish bank (Piccoli et al., 2022).

We posit the *strategic primacy* of digital resources as the minimum necessary condition for a firm to be classified as digital. In such a scenario, the strategic initiatives that define the firm's value proposition and/or its competitive position leverage digital resources as a necessary element in their execution.¹³ This minimum necessary condition requirement offers theoretical and pragmatic benefits. First, it is industry and scale agnostic. It applies equally well to “born digital” ventures like Netflix, century-old stalwarts like heavy industrial equipment leader Caterpillar, and government institutions like the Australian Tax Office. Second, it does not require attention to the business model (Wessel et al., 2021), and it does not discriminate between business-to-consumer, business-to-business, or non-profit institutions. To determine whether an organization is architected as digital, considerations like the nature of the customer, the firm's value proposition, or the organization type are irrelevant. Third, it identifies a structural difference – the architectural commitment to digital resources – that is both durable and prodromic to organizational transformation requirements. Durable because it is not tied to “modern” waves of digital technology but rather to the organization's digital architecture and the structure of the enduring organizational resources underpinning the company's strategy (i.e., it is irrelevant if a digital resource uses relational

¹¹ The potential for assets and capabilities to be rearchitected as digital resources is continually expanded by digital technology's progress. Thus, a digital firm is constantly reevaluating resources, looking for external ones to orchestrate, and assessing the feasibility of rearchitecting internal assets and capabilities as digital.

¹² MACH stands for Microservices, API-first, Cloud-native and Headless. Composability is the term used by Gartner to refer to applications that abandon the “proprietary technology stack of the ‘monolithic approach’ that [traditional] ERP vendors deliver” (Faith et al., 2020, p. 3) in favor of modular solutions enabling the foundational administrative and operational digital capabilities of the organization to support assembly and reassembly of business processes and application experiences.

¹³ *Necessary* here implies that executing the initiative without digital resources is unfeasible or impossible. As a corollary, it also indicates that the initiative or value proposition would not be the same if the digital resources at its core were swapped for IT-enabled or complementary substitutes, which is why the firm elects to orchestrate digital resources.

databases or blockchain as its core data management technology). Prodromic because the strategic choice to embrace the primacy of digital resources precipitates changes in organizational deep structures, requiring a sweeping transformation process. Finally, the definition of digital organization is ontological. Its contribution is to enable rigorous identification of this organizational state, thus allowing for theory development about its genesis (e.g., as a “born digital” organization or through digital transformation), its impact (e.g., on firm agility or execution speed), and its contingencies (e.g., if, when, and how an organization should become digital).

It follows that embracing the primacy of digital resources is an explicit strategic choice that may or may not be appropriate for any one organization at a specific point in time. It also does not mean that all digital resources are inherently valuable. Rather, it means that the firm recognizes and leverages digital resources as “core [to its] (potentially flexible and adaptive) intended course of action” (Van den Steen, 2017, p. 2616) such that it makes a strategic choice to prioritize the use of digital resources in its resource bundle, as exemplified by the following quote:

“We [at Instacart] tried to build the things that we are uniquely suited to build...we are uniquely suited to build a great front-end ecommerce experience, and a great enterprise experience for grocers, we are uniquely able to fulfill, we are uniquely able to onboard shoppers and give them a great experience. We are not as well suited as other people [like Stripe] to have the payment infrastructure for the entire thing.” (Gupta, 2019).

Instacart leadership made the strategic choice to focus development on a few uniquely-defining elements of its value proposition – orchestrating all other components from external digital resource providers like Stripe for payment, Twilio for communication, and AWS for storage and compute (Piccoli et al., 2022).

Digital transformation

Embracing the strategic primacy of digital resources makes digital organizations architecturally and structurally different from IT-enabled organizations. The definition provides rigorous anchor points for understanding the metamorphic process of digital transformation by providing clarity about the beginning and end state of such a process (Markus & Rowe, 2023). While firms engaged in multi-year transformation journeys may intermediately display traits of both (Haskamp et al., 2022), delineating the nature of the beginning and end states provides the conceptual schema for identifying and evaluating the change. Moreover, since a digital organization is structurally different from its IT-enabled counterparts, we can delineate the unique traits of digital transformation and, like adding a new vanishing point to a perspective sketch, shift the focus of future research toward previously hidden questions. The outcome of digital transformation is a digital organization – one committed to the strategic primacy of digital resources in its core guidance as reflected in its digital architecture and design principles. Thus, this end state is not a place of “quiet stability ever after.” Rather, with the change in digital technology architecture, the organization commits to orchestrating digital resources when available, creating them when not available, and employing non-digital organizational resources when necessary. The completion of a digital transformation journey (i.e., the arrival point of the organizational metamorphosis) does not bring stasis; rather, it is the beginning state of expanding digital possibilities, and the organization will continue to evolve and innovate. In some cases, evolution might be minimal. But in competitive industries, the digital organization will be characterized by a “constant turbulent quest for new digital capabilities and digital assets”¹⁴ in the increasingly combinatorial, servitized, and infrastructural environment that characterizes the modern competitive landscape (Piccoli et al., 2022).

Critics may argue that our view is restrictive, that characterizing the digital organization based on the role of digital resources is arbitrary and technocentric. However, the literature’s persistent inability to clearly articulate the difference between digital transformation and IT-enabled transformation (which organizations have been engaging in since the days of business process reengineering) (Carroll et al., 2023), suggests that sharper demarcations and a change of perspective are needed (Markus & Rowe, 2023). Simply because the term digital transformation has entered the business and consulting vernacular is no justification for scholars to uncritically adopt it. IS academics are often accused of “journalism” (Watson & Seidel, 2018): of documenting or describing technological changes that were pioneered by practicing managers. To avoid such a trap, we should strive to engage in theoretical prescience: “the process of discerning what we need to know and *influencing the intellectual framing* [emphasis added] of what we need to know” (Corley & Gioia, 2011, p. 23). Our framing is the primacy of digital resources and the unique ontology of digital organizations.

Furthermore, we note that attention to the structural differences between IT-enabled and digital organizations does not imply a technocentric or materialistic view. Rather, it provides a conceptual anchor point to identify material differences, not only in technology architecture and design principles but also in the processes, people, structure, and culture that fit the novel architecture and principles. Early research and previous theory point to some of these differences (see Table 2 for an illustrative list). For technology infrastructure, the transformation requires architectural flexibility (Mikalef et al., 2021) through the reconfiguration of core assets and capabilities into resources that enable digital resource creation, orchestration, and management. Examples include the introduction of new roles, such as digital resource creators, curators, and composers (Thomas & Dumas, 2022), as well as the upskilling of developers in cloud-first techniques and the rise to prominence of digital product managers over traditional project managers (Nelson, 2024). Organizational processes evolve from tightly coupled procedures that support long-range projects to iterative product development processes grounded in the configurations of resources, often architected on digital platforms (Haskamp et al., 2022). For organizational structure and culture, this involves the governance of digital resources and a mindset of co-creation of value facilitated by the fluid nature of digital material (Grover et al., 2024) that may result in organizational identity changes (Wessel et al., 2021). For example,

¹⁴ We thank the Senior Editor for suggesting the clarifying wording quoted in this paragraph.

Table 2

IT-enabled organizations versus digital organizations characteristics.

	IT-enabled Organization	Digital Organization
Technology Infrastructure	· Integrated monolithic applications from vendors · Proprietary data centers and application outsourcing · Adoption of technology stacks (e.g., Hadoop) rather than digital resources · Static Extract, Transform, and Load (ETL) processes	· Digital architecture enforcing coherent design principles – e.g., extensibility, and backward compatibility · Reusable, composable, fully formed digital resources (internal/external) · Cloud-first deployments and re-platforming · Event-driven architecture and digital data streaming
People	· System developers · IT professionals and internal business clients · Project managers · Traditional mindset, compliance-oriented Chief Information Officers (CIO)	· Cloud-first developers · Digital resources creators, curators, and orchestrators · Product managers and digitally savvy citizen developers · Agility oriented CIOs, Chief Digital Officers (CDO) and digitally savvy executives
Processes	· IT alignment · Business process automation/reengineering · Staged innovation process · Planned use cases and lengthy, formal justification for projects	· Digital resource creation, curation, and provision · Maintenance of digital resources as products designed for reuse and recombination as modules through cataloging, deprecation, documentation, testing, etc. · Digital resources orchestration and experimentation (problem–solution spaces)
Structure and Culture	· Compliance-driven enterprise architecture · Positional power · IT project orientation · Product lines and departments · Bureaucracy and ad-hoc coordination · Separation between IT and internal business clients	· Agile enterprise architecture · Orchestrate first, create second; adopt digital resources when they become available · Business units as “product owners” of digital resources · Foster domain knowledge by fusing and aligning deep business knowledge with technical competencies · Ecosystem co-creation and pervasive digital culture

organizations recognize that traditional enterprise architecture practices emphasize compliance and IT control but often stifle execution speed and responsiveness. The agile enterprise architecture model, conversely, is “business outcome-driven, service-centric, nimble” (Muthukaruppan & Nandey, 2019, p. 5). Firms might experience the inherent clash between leveraging digital resources in “multiple value paths, offered through design recombination” (Henfridsson et al., 2018, p. 89) and core tenets of agile software development because “to design for reuse is counter to the motivations of agile methodologies unless they have defined user stories calling for that reuse” (Thomas & Dumas, 2022, p. 3).

Describing digital transformation as a change of state implies that those organizations that are architected as digital at their inception (i.e., born digital) will not need to transform. Conversely, those currently designed as IT-enabled organizations may elect (or not) to embrace the strategic primacy of digital resources. We argue that it is because of the lack of precision in defining the unique nature of digital transformation that IS scholars have yet to systematically uncover the defining technical, organizational, cultural, and talent enablers of successful digital transformation (Markus & Rowe, 2023). This gap provides significant opportunities to contribute knowledge in this area based on a common baseline understanding of digital transformation and digital organizations.

Research implications

We have alluded to two ontological grounds for distinguishing digital transformation from IT-enabled transformation: the *ontological unit* and *ontological reversal*. Both have profound implications for research. For the former, traditionally, we characterize IT units as boxes of hardware and software. While IT boxes still exist in digital organizations, they fade to the background, in most cases becoming the run-time material bearer of digital resources (Faulkner & Runde, 2019; Lyytinen, 2022). Thus, the central ontological unit is no longer the IT artifact as traditionally conceptualized (Orlikowski & Iacono, 2001). Rather, it is the digital resource at the heart of the firm’s digital infrastructure and value creation paths. The ontological distinction of the unit has significant implications for digital organizing; instead of having social manifestations at the higher level of hardware and software boxes, the semantic links are at a far more granular level in the digital resources. This approach explicitly acknowledges and models the affordances of digital resources (e.g., modularity, programmatic access) that enable flexibility in orchestrating internal strategic initiatives (e.g., preventive maintenance) and external value propositions (e.g., two-hour grocery delivery). For the latter (i.e., ontological reversal), the primacy of digital resources is paramount since “the nonphysical digital version of the reality is not just as real as the physical version, it is more so” (Baskerville et al., 2020, p. 511). By giving primacy to digital resources, which regardless of their internal structure are exposed to the external world as software modules (Piccoli et al., 2022), the organization recognizes that these digital objects come first, at least with respect to strategic and value creation initiatives. Such recognition, and the ensuing transformation, enables the firm to compete and operate as a “software company” (Gnanasambandam et al., 2022) regardless of the nature of its product, service, or industry.

The implications of these ontological distinctions should change our research conduct.¹⁵ Consider the extrapolation of existing research. If digital transformation is conceived as ontologically equivalent to IT-enabled transformation, it will deal with the use of new waves of digital technology to enable deep structure change that fosters new value propositions, enhances products and services, leverages data, and creates new revenue models and customer experiences. The questions addressed are important but largely deduced from our accumulated “legacy” knowledge of the mechanisms of process redesign, automation, cultural change, and organizational redesign. However, in the spirit of Alan Kay’s notion that “point of view is worth eighty IQ points” (Kay, 2021), making the ontological distinction in conceptualizing digital transformation provides researchers with a vantage point that exposes a whole new set of “hidden” questions that are critically important for our digital future. For example, regarding questions of expected outcomes, the literature has yet to answer whether a digital transformation is a requirement or a beneficial endeavor for all organizations and how executives should think about the risk/reward tradeoff entailed by the digital transformation journey.

Despite the observation that software increasingly “programs the world” (Andreessen et al., 2016) and consulting firms’ advocate for operating models “where technology and business collaborate on developing and delivering digital products and services” (Lamarre et al., 2021, p. 2), there is no conclusive evidence that all firms must become digital organizations to be successful. The literature identifies many of the expected benefits of successful digital transformation, increased agility, option value, and execution speed (Benzell et al., 2022; Salmela et al., 2022; Woodard et al., 2013), but the contingencies, risks, and costs have yet to be understood. More generally, unanswered questions revolve around the digital organization and its digital architecture, as well as its processes, people, and culture. There are questions pertaining to the process, such as the stages and metrics we should use to evaluate and monitor the digital transformation process as it unfolds. Other questions refer to the precipitating conditions – when is digital transformation necessary? Our hope is that the conceptual elements introduced here, the primacy of digital resources and the digital organization, will push future research in this area in a more coherent direction. Below, we point to examples of fruitful research opportunities driving an initial research agenda around questions of strategy, scope, and ongoing change, as well as data and infrastructure.

Questions of strategy

The strategy literature identifies three logics to describe how firms attain superior performance: positioning, leverage, and opportunity. The logic of positioning is predicated on recognition of structural industry characteristics (e.g., 5-force model) and the firm’s ability to use digital technology to situate itself advantageously in an extended network of suppliers and customers. The logic of

¹⁵ Until very recently, most IS research has been concerned with boxed units being effectively adopted, used, and deployed innovatively in organizations. The primacy of digital resources demands consideration of infinite configurations and emergent value pathways.

leverage is predicated on control of valuable, rare, immobile, and non-substitutable (VRIN) resources that may be heterogeneously distributed among competitors (Barney, 1991; Nevo & Wade, 2010). Firms can achieve sustained competitive advantage by developing custom IT resources (Piccoli & Ives, 2005) or integrating digital technology resources with complementary organizational resources that create IT-enabled resources with emergent properties (Nevo & Wade, 2010). Finally, the logic of opportunity is predicated on the ongoing exploitation of prospects through competitive actions grounded in sense and respond activities as well as dynamic capabilities (Rindova & Kotha, 2001), enabling continuous learning and reconfiguring processes that allow the firm to continuously navigate short-lived competitive advantages (McGrath, 2013). Despite the occasional (misguided) critique that “IT doesn’t matter” (Carr, 2003), the field has extensively demonstrated that digital technology investments enable value creation and appropriation of value, explaining the findings with one of the above logics.

It is an open question how, after successful digital transformation, a digital organization may attain continued superior performance. What elements of established models for analyzing IT-enabled competitive advantage (e.g., Piccoli & Ives, 2005) remain applicable to digital organizations? Previous theoretical work suggests that innovations in digital technology architecture leading to “modularized productive means and standardized interfaces” can promote turbulent competitive environments (Nan & Tanriverdi, 2017, p. 943). This occurs by enabling a wider array of resource combinations and simultaneously increasing the complexity of identifying those bundles that yield superior performance. Digital resources partially epitomize this theoretical framework by modularizing assets or capabilities and making them programmatically available for reuse and recombination. Following such theoretical work, we anticipate that organizations undergoing digital transformation will enhance their ability to experiment with a broader spectrum of resource bundles, drawing both from digital resources internal and external to the organization. While we can expect that digital organizations are likely to outperform IT-enabled organizations, at least on the dimensions of agility, responsiveness, and speed to market, there is currently a lack of empirical research to substantiate and theoretically justify this speculation. More generally, we need to investigate whether existing theoretical explanations can elucidate variations in performance among digital firms as well as between digital and IT-enabled organizations.

Furthermore, the context could matter. It remains unclear in which specific contexts (e.g., hypercompetitive environments) and for how long digital firms might outperform their IT-enabled counterparts. With the increasing number of organizations embarking on digital transformation, the competitive landscape is likely to become more dynamic and intensely competitive. A promising approach to investigate the above questions is to understand how intermediate outcomes of digital transformation – such as enhanced agility, increased flexibility, and business process performance – contribute to the sustained potential for value appropriation (Baiyere et al., 2020; Salmela et al., 2022). In addition, given that the architecture of digital firms alone may not be adequate for seizing opportunities and responding swiftly to evolving market conditions, it is essential to explore and understand the moderators and mediators influencing this relationship. For example, we must understand the role of organizational capabilities, such as sensing capabilities (Overby et al., 2006) or the ability to renew the portfolio of IT applications (Queiroz et al., 2018), in moderating the agility of digital organizations. To effectively manage the available digital resource bundle, organizations must discern when to develop new digital resources, utilize existing digital resources, or retire outdated resources in favor of digital assets or capabilities.

After successful transformation, the primacy of digital resources implies that modular, bitstring-encapsulated assets and capabilities comprise a significant portion of the firm’s bundle of resources. Recent work contributes to our understanding of how to architect digital resources to optimize their adoption by external firms (Sandberg et al., 2020). But how does the digital firm ensure that such resources are harnessed for superior performance? We are in the nascent stages of this exploration (Helfat et al., 2023), and while the logic of leverage would appear to be the most suitable theoretical framework, the unique nature of digital resources requires empirical testing. There is early evidence that “inverting the firm” (Parker et al., 2017) by crafting and exposing digital resources that underpin complementors’ value proposition is a viable strategy for digital organizations (Benzell et al., 2022). Recent mathematical modeling work illustrates how the unique characteristics of digital firms sets them on a path to achieve both high specialization and hyperscale (Giustiziero et al., 2023). While formal models are compelling, testing and validation is needed. More importantly, several contingencies remain undertheorized. For instance, certain firms opt for orchestration of external digital resources rather than in-house creation (Piccoli et al., 2022). How does ownership versus contractual control of external digital resources by digital organizations change the potential for value capture? While it is appealing for digital transformation journeys to rely heavily on the use of digital resources on the public cloud, what are the future performance implications of such choice? We are studying a large manufacturing company that decided to develop some digital resources in house, accepting a slower and more costly transition toward the digital organization end state because of the executives’ assessment that such resources were strategic and had to be controlled. Similarly, Dropbox transitioned away from digital resources offered by public cloud providers to develop their own custom infrastructures and technology stack with the objective of achieving economies of scale and reducing costs. We have no theoretical framework to evaluate the accuracy of such decisions and, consequently, to provide guidance to managers. Such a framework would have to include an explanation of the value capture potential of digital resources that accounts for their unique ontology (Grover et al., 2024).

Similarly, we have no clear guidance as to the question of co-specialization of digital resources. External digital resources, by definition, are readily available in the market – from cloud vendors like AWS to vertically specialized resource providers like Stripe or Twilio. Because they are modular components, not tailored or co-specialized to the idiosyncratic characteristics of any one specific firm, it is unclear the extent to which widely available digital resources can serve as a basis for differentiation. Recent work points to the importance of architectural capabilities as “an important subset of dynamic capabilities that provide managers with the ability to see a complex technical system in an abstract way and change the system’s structure by rearranging its components” (Baldwin, 2015, p. 2). The capacity to envision where to stake a valuable position in an emerging complex technical system is most likely the core element of a successful orchestration strategy, as evident in the strategic choice Apple made when committing to HTML5 and foregoing Adobe Flash support to avoid “a third-party layer of software coming between the platform and the developer” (Jobs, 2010). Empirical

validation of this and similar propositions provides fertile ground for future research as more and more digital organizations engage in digital resource orchestration. The furious pace of work going on in AI automation, where intelligent agents are orchestrated through APIs to accomplish increasingly complex tasks, is a current example of the importance of architectural capabilities and an apt context for future research.

Co-specialization is a critical prerequisite for capitalizing on resources utilized by others (Dyer et al., 2018; Dyer & Singh, 1998), yet it is possible that the orchestrated resource bundle, along with a few unique internally developed capabilities, engenders emergent properties that lead to sustainable advantages (Nevo & Wade, 2010). Digital resources are not homogeneous, and different digital organizations will display variation in their bundles of digital resources. Thus, future research needs to move beyond sweeping generalizations about the bundle compositions, such as the assumption that all digital resources are inherently scalable (Giustiziero et al., 2023). The substantial losses incurred by digital organizations that operate in physical space when they attempt to scale their services (e.g., Uber, DoorDash) provide anecdotal evidence that hyper-scaling may not be straightforward, even for the digital organization. Moreover, questions of complementarity remain open as there is no research that we are aware of investigating the synergy of traditional organizational assets and capabilities with digital resources.

Finally, the core guidance view of strategy pays particular attention to the role of the strategist (Van den Steen, 2017). Embracing the primacy of digital resources implies a mindset change for the organization, but whether that is prodromic or consequent to the digital transformation remains to be discovered. Moreover, a clear definition of the nature of digital strategy and the role of architecture and design principles at its core begs the question of what kind of strategist will be best suited to initiate a digital transformation and see it through to completion. There is a growing literature on the characteristics of CDOs and how their role differs from that of traditional CIOs. Early findings indicate that top managers already employed cannot handle digital transformation as an added responsibility (Tumbas et al., 2018). More importantly, the proliferation of C-level positions begs the question of whether the management of digital transformation requires specialized skills and spheres of influence (Lorenz & Buchwald, 2023; Sciuk & Hess, 2024). Is the addition needed because the sheer amount of work cannot be handled by the CIO or CEO? Is it because the managerial skillset is unique and not typically possessed by the conventional CIO or CEO? The unique ontology of digital transformation hints to the possibility that there are special idiosyncrasies to the CDO position, idiosyncrasies that are still little understood, as attested to by empirical evidence of the relative short tenures of CDOs (Firk et al., 2021).

Questions of scope and ongoing change

Panta rhei, the Greek aphorism meaning “everything flows,” remind us that the only constant in (organizational) life is change. Our framing of digital transformation as a metamorphosis suggests a discrete, one-time change of state. A legitimate question is then what happens upon completion of the digital transformation process? While there is research addressing the need to “sustain” or see through the complex endeavor that is a digital transformation *process* (Carroll et al., 2023), there are no studies that we are aware of rigorously answering the question of repeated digital transformations. Once the firm has committed to the primacy of digital resources, or for organizations “born digital,” how should we qualify inevitable future deep structures changes?

Our position is that a digital transformation is a one-time occurrence. Once the metamorphosis from IT-enabled to digital organization is complete, the same entity will not engage in another *digital* transformation.¹⁶ It may, of course, undergo other types of transformations (see below). This contention is based on the core element of digital transformation as we define it: a change in digital technology architecture rather than a change from digital technology use. The pivotal aspect of the transformation is the firm’s strategic choice (Van den Steen, 2017) to commit to digital resources as the core element of its digital technology architecture and to engage in complementary deep structure and organizational change. Upon completion of the digital transformation process, the firm has morphed into a digital organization. While new digital resources will become available and constantly be developed as business needs evolve and opportunities emerge, the architecture will remain largely stable. As an example, consider Caterpillar, the century-old industrial firm that recently rearchitected data operations around a cloud data platform called Helios which is designed to create “the ultimate source of truth for all our applications” as the key enabler of reliable digital services (The *Industrialist*, 2023, p. 3). Such transformation effort requires a clear set of architectural decisions and design principles. It necessitates clear core guidance and strategic commitment to ensure that complementary organizational, process, and competency changes are implemented. The primacy of digital resources has been established, and the architectural transition has indeed occurred, thereby signaling the successful completion of the transformation. Caterpillar is constantly looking for new data to rearchitect into the Helios platform and for opportunities to leverage externally created digital resources to substitute for those that it had to internally create. The firm is by no means in a state of “quiet stability ever after,” but these evolutionary changes do not qualify as digital transformations. Competition continues relentlessly, and the firm pursues new opportunities when available; however, the metamorphosis has taken place.

Note that the question of repeated transformation is not unique to *digital* transformation. Rather, it is a question about the boundaries of any organizational transformation. Based on observations of many digital organizations, we posit that subsequent organizational transformations will be characterized by novel catalysts. For example, a digital organization may engage in business transformation if it radically changes its products or business model. While the underlying digital architecture of the organization will not change, the company may need to acquire new competencies or change its identity. An example of such business transformation is currently offered by car manufacturers under the European mandate to switch from internal combustion to electric drive trains and

¹⁶ Theoretically, a second digital transformation would be possible if the firm first reverts to an IT-enabled organization form, and subsequently reconfigures as a digital organization – practically unlikely, but theoretically possible.

batteries as the core technology in their products. Irrespective of whether they embrace the primacy of digital resources, these companies must radically change their product lines and core offerings. Yet as they attempt to radically transform their value propositions, they struggle to maintain the relevance of their existing competencies (Liu et al., 2024).

But what about future *technology* transformations? If there cannot be multiple digital transformations, will there be future technology transformations? We suggest that the question should be addressed with the same ontological precision we adopted in this article. Consider quantum computing, which does not rely on digital technology since quantum computers are not digital computers. What would qualify as a “quantum transformation?” If quantum computing resources are modularized and encapsulated in a programmatic bitstring interface (i.e., they configure as digital resources) and they are incorporated in the existing digital architecture and design principles of digital organizations, we see no need for transformation. There will be evolution, with a likely quest for new assets and capabilities, perhaps resembling the current race to incorporate generative artificial intelligence in existing value propositions (Brynjolfsson et al., 2023). But the primacy of digital resources will endure with no need for organizational transformation. Conversely, if the quantum computing paradigm shifts the architecture of the computing resources, requiring a reevaluation of the primacy of digital resources, we hypothesize that digital organizations will engage in “quantum transformation” and a metamorphosis will need to occur from digital organization to “quantum organization.”¹⁷ As with digital transformation, we would expect such an organizational transformation to result in material differences, not only in technology architecture and design principles but also in the processes, people, structure, and culture that fit the novel design.

We have developed our arguments using the organization as the level of analysis and the focus of transformation. In its archetypal form, an IT-enabled organization devises a business strategy and, through IT alignment, ensures that its digital technology strategy fits with “the business” (Baskerville et al., 2020). Conversely, the prototypical digital organization recognizes digital resources as core to its intended course of action, guiding lower-level organizational decisions toward the systematic creation, reuse, and recombination of digital resources. A digital organization devises processes, recruits and retains employees, and implements organizational structures designed to foster agility and execution speed through the reuse and recombination of internally developed and/or externally procured digital resources. Within this view of digital transformation, many novel questions await exploration. For example, while an entity will undergo only one digital transformation in its existence, such an entity is not necessarily a company. An intriguing area for future research is the boundary of a digital transformation in terms of business units, organizational functions, or focus. Specifically, is it possible for an organization to experience the substantial architectural change we describe here in one business unit and not in others? Is it possible for the organization to experience the metamorphosis only in one or two business functions? How should we conceptualize a bank that acquires a “born digital” fintech but keeps its operations separate? Anecdotal evidence suggests that the impetus for digital transformation often stems from one or more business units, or functional areas, that have a particular interest in leveraging digital resources. The emergence of modular business architecture (Greeven et al., 2021) as well as consistently lowering barriers to self-provisioning of software via SaaS solutions (Lawson, 2021) or low-code/no-code development (Bucaioni et al., 2022; Novales & Mancha, 2023) also suggest a positive answer. Because digital transformation stems from the strategic choices the firm makes, as captured in its core guidance, we suspect that the answer largely depends on the organizational structure and the degree of strategic independence of the units. Yet, rigorous analysis is needed to explore these questions. Related inquiries include the conditions under which these “local” digital transformations succeed or fail and the wisdom of creating separate “digital units” in organizations in the process of digital transformation, as successfully done at BBVA (Piccoli et al., 2022). As the CDO of a Fortune 100 company recently told us: “I think my job should not exist 10 years from now.” The statement referred to his belief that the digital organization critical role was in navigating a transformation toward a new architecture. But if the proliferating digital units being established in organizations are to be transitory, what is the appropriate organizational governance structure to curate, manage, and develop core digital resources within the established architecture? Can the IT function reassert its ownership of such tasks upon completion of the digital transformation process? Conversely, if digital units are not transitory, under what circumstances can they coexist with IT-enabled business units?

Questions of data and infrastructure

The concept of infrastructure has received research attention for more than three decades. Building a responsive IT infrastructure – defined as the “physical assets, intellectual assets, and standards” that serve as the foundation of organizational information systems (Chatterjee et al., 2002, p. 10) – was a key IS management issue in the 1990 s (Niederman et al., 1991). While all infrastructural hardware and software assets are important, data is increasingly seen as a strategic resource (Piccoli & Ives, 2005; Wixom et al., 2021). Traditionally, companies have created value from data by supporting strategic business processes (Chen et al., 2012), improving their decisions when responding to the competitive environment (Rainer & Watson, 1995), or increasing the efficiency of their operations (Melville et al., 2004). More recently, technological advancements – such as cloud computing, multicloud technology, and shared data architectures – facilitate the emergence of novel data monetization strategies (Grover et al., 2018; Wixom et al., 2023). Yet, the ability to leverage data for value-creating activities is contingent on the degree to which data assets are structured for reuse and recombination across the organization (Wixom & Piccoli, 2020).

While previous research on IT infrastructure flexibility (Byrd & Turner, 2000) and data warehousing (Wixom & Watson, 2001) recognizes the importance of improving the retrieval and flow of data within and across organizations, it is only recently that we have

¹⁷ The label “quantum transformation” and “quantum organization” are placeholders; at this stage, we have no insight into what the nature of such transformation, or the appropriate label to distinguish it, might be.

started to investigate how infrastructures decouple software, hardware, and data components (Mikalef et al., 2021; Tiwana et al., 2010). The bundle of digital data assets available within digital organizations plays a vital role in realizing such vision by facilitating the reuse and recombination of data for a range of anticipated and unforeseen future applications (Piccoli et al., 2022; Wixom et al., 2021).

Despite recognizing data as a central resource for enabling organizational change (Alaimo & Kallinikos, 2022) and general statements asserting that “what digital transformation means for strategy” requires “an understanding of the characteristics of data as a strategic resource” (Adner et al., 2019, p. 256), there is a paucity of research investigating the unique characteristics of data-driven digital transformation. Given the pervasiveness of data across organizational processes and its importance for developing value propositions, investigating the role of digital data assets as a catalyst or driver for digital transformation offers fertile ground for future inquiry. While structuring organizational data as digital resources seems straightforward at first glance, numerous aspects warrant deeper exploration. In fact, while acknowledging the primacy of digital resources, digital infrastructures can configure in multiple ways (Henfridsson & Bygstad, 2013). Historically, data within IT-enabled organizations have resided in application silos or centralized databases and were not intended for reuse and recombination across the organization. Moreover, as we develop theories to explain the transformations data undergo from generation to contextualization for establishing their value (Aaltonen et al., 2021), it is imperative to enhance our understanding of the specific processes underpinning the effective management of digital data assets and the specific organizational structures they need in support. While we are beginning to gather insights into digital infrastructures with a focus on the role of data (Tremblay et al., 2023), we have rudimentary understanding of the specific requirements, challenges, and mechanisms necessary for the development, reuse, and recombination of digital data assets.

Crucially, with data emerging as a central resource in numerous digital products and customer solutions, there is a growing trend of rearchitecting traditional data assets into digital data assets, enhancing their utility and value across the organization (Wixom & Beath, 2022). Leaders in this architectural transition advocate for a fundamental shift in data reuse and recombination methods. They propose moving away from traditional technologies such as Extract, Transform, Load (ETL), File Transfer Protocol (FTP), and even Application Programming Interfaces (APIs), towards scalable cloud data platforms (Miller, 2018). These data platforms provide access to always current views by using streaming data services and do not require data transformation or duplication when sharing. Despite the undeniable commercial success of several data platform solutions, we don’t have a clear understanding of the tradeoffs that different digital data assets architectures introduce (e.g., APIs versus event streaming).

It’s also crucial to acknowledge that not all organizational data assets will be architected as digital data assets. This realization presents new research opportunities, such as understanding how the architecture of digital data assets can coexist and interface with legacy infrastructure within an organization, as integration will pose challenges. Anecdotal evidence suggests that some companies, unwilling to modernize parts of their infrastructure, are encapsulating legacy systems by building new abstraction layers that increase their modularity and make them accessible programmatically (Van Der Zijden & Dodd, 2023). Since most companies are only at the beginning of their transition to modern cloud data platforms, it is not yet clear what the benefits and challenges are. Furthermore, this situation prompts an exploration of the types of data assets that should or should not undergo rearchitecting. Recent research advocates giving precedence to strategic data assets in rearchitecting efforts (Wixom et al., 2021), but our theoretical understanding of these evaluations is still in its infancy.

Finally, there are questions related to centralization and decentralization of data infrastructure and management. While the discourse on centralization versus decentralization in data management is longstanding, modern data platforms enable for the first time the creation of global, scalable organizational platforms, a concept that was previously only theoretical. In fact, while several technologies, such as Hadoop, promised to facilitate such unified architecture, their practical applications often left large organizations to implement multiple data platforms tailored to specific regions, business units, and use cases. While technological advancements make global data platforms feasible, the governance of digital data assets remains a complex and under-researched area. Balancing a global data infrastructure might necessitate decentralized management for digital data assets. In fact, while the technical expertise needed to deploy modern data platforms could be centralized, the domain-specific knowledge necessary for curating data and ensuring its quality and relevance across various use cases often resides within the business units that generate the data (Alaimo & Kallinikos, 2022). Case evidence suggests that digital organizations may require new organizational structures (e.g., data boards), skills (e.g., domain data owners), and processes (e.g., data certification programs) to resolve such tensions and ensure the desired reuse and recombination of digital data assets. Observations from large multinational organizations indicate that managing these data platforms may necessitate the creation of dedicated organizational units integrating cutting-edge technical capabilities with in-depth data knowledge, serving as liaisons with data owners throughout the organization (Caterpillar, 2022). As an increasing number of companies embark on digital transformation journeys, they provide fertile ground for understanding this unique phenomenon and formulate rigorous theory.

Note that the questions raised in this section implicitly reiterate the case for ontological precision. They can only be framed once we have clear definitions and boundaries to complex concepts, like digital transformation, that delineate the unique nature of the phenomenon. Digital organizations, those embracing the primacy of digital resources, are the ones that are digitally transformed. Metaphorically, our attention shifts from “mixtures” of digital technology and organizational resources to digital “compounds” exhibiting new properties not reducible to the digital technology or the organizational resources from which they stem.

Conclusion

In this paper, we argue that a useful way to conceptualize digital transformation is as the metamorphosis of an organization from one that is, to varying degrees, enabled by digital technology to a digital organization. We conceptualize this change through two

related concepts: digital resources – the class of digital objects that are modular – encapsulate objects of value and are accessible by way of a programmatic interface; and a digital organization, one that embraces the creation, reuse, and recombination of digital resources as a primary driver of its value proposition and internal operations. Our work is unique in that we characterize digital transformation as a change *in* digital technology architecture rather than a change *from* digital technology use.

While we acknowledge the complexity of digital transformation journeys and the importance of rigorously studying the digital transformation process, we maintain a disciplined focus on the ontological definition of digital transformation. Our emphasis is on the organization and the nature of the change (i.e., the ontology of digital transformation as an organizational metamorphosis). Failing to clearly articulate the properties of organizations pre- and post-transformation and failing to identify the unique “mode of being” (Faulkner & Runde, 2019, p. 1283) of a digital organization as compared to IT-enabled organizations makes it difficult to point research attention toward timely and relevant research questions. Empirical inquiries like the ones delineated above benefit from clear definitions (Markus & Rowe, 2023), leading to a body of literature that is more consistent and rigorous. With a focus on the ontological foundations of digital transformation, we offer a contribution to basic science, bringing definitional precision to labels that have gained currency in academic and managerial discourse. We also hope that this article serves the purpose of knowledge accumulation that defines the academic endeavor, while helping to place information systems research at the center of digital strategy theorizing and raise its profile as a reference discipline for adjacent scholarly fields.

CRedit authorship contribution statement

Gabriele Piccoli: Conceptualization, Lead drafting and writing. **Varun Grover:** Conceptualization. **Joaquin Rodriguez:** Literature review over, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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